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|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>EDUCATION</b>              | <b>Ph.D. in Computational Science</b> (Interdisciplinary Program combining Science, Engineering, Computation, and Mathematics)<br>The University of Texas at El Paso (UTEP), El Paso, TX                                                                                                                                                                                                                                                                                                                                                                                                                           | Awarded: Dec 2024   |
|                               | <b>Master of Science in Physics</b><br>UTEP, El Paso, TX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Awarded: Dec 2019   |
|                               | Tribhuvan University (TU), Kathmandu, Nepal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Awarded: Dec 2016   |
|                               | <b>Bachelor of Science in Physics</b><br>TU, Kathmandu, Nepal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Awarded: Jan 2013   |
| <b>PROFESSIONAL POSITIONS</b> | <i>Professor of Science - Physics</i><br>Department of Science and Agriculture<br>Central Texas College (CTC), Killeen, TX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Sep 2025 - Present  |
| <b>ACADEMIC EXPERIENCES</b>   | <i>Research Scientist</i><br>Department of Physics, UTEP<br>El Paso, TX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Jan 2025 - Dec 2025 |
|                               | <ul style="list-style-type: none"><li>• Investigated lattice dynamics, phonon behavior, and thermal properties of metals and alloys under a wide range of temperature and pressure conditions using atomistic simulations.</li><li>• Modeled free energies, phase stability, and thermodynamic properties of materials beyond the quasiharmonic approximation, incorporating anharmonic effects.</li><li>• Developed and applied advanced computational methods, including Molecular Dynamics and machine-learning-based interatomic potentials, to study materials with near first-principles accuracy.</li></ul> |                     |
|                               | <i>Graduate Teaching Assistant</i><br>Department of Physics, UTEP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Aug 2024 - Dec 2024 |
|                               | <ul style="list-style-type: none"><li>• Physics Instructor - Conducted workshop (three sessions per week) for undergraduate Introductory Mechanics course (including teaching, designing quizzes, &amp; grading).</li></ul>                                                                                                                                                                                                                                                                                                                                                                                        |                     |
|                               | <i>Graduate Research Associate</i><br>Department of Physics, UTEP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Aug 2023 - Aug 2024 |
|                               | <ul style="list-style-type: none"><li>• Used classical (LAMMPS) &amp; Quantum Molecular Dynamics (QMD) calculations of vibrational spectra of solids as a function of temperature.</li><li>• Studied material behavior using <i>ab-initio</i> calculation (QMD, DFT, VASP, Quantum Espresso, QHA).</li><li>• Reproduced the accurate thermal pressure and equation of states (EOS) in transition metals by including the effect of temperature on the phonon density of state (DOS).</li></ul>                                                                                                                     |                     |

*Visiting Summer Research Student*  
The University of California at Berkeley

Jun 2023 - Jul 2023

- Studied the magnetic Order-Dependent Properties of FeV and Fe<sub>3</sub>V Alloys using density functional theory (VASP).
- Built HPE workflows for material modelling using Python, Bash environment.
- Machine learning assisted analysis of material datasets.

*Graduate Teaching Assistant*  
Computational Science Program, UTEP

Aug 2019 - May 2023

- Teaching Assistant - Assisted professors with grading papers, & conduct weekly workshops for assigned undergraduate/graduate Computer Science (Statistical Data Mining (using R and Python) and Statistical Programming in R) courses.
- Math Tutor - Assisted students in undergraduate Math classes (Discrete Mathematics, Differential Equations, Matrix Algebra, & Calculus - up to Calculus III) including concept clarification & problem-solving strategies.

*Graduate Teaching Assistant*  
Department of Physics, UTEP

Aug 2017 - July 2019

- Taught undergraduate Introductory Physics (Electromagnetism, Introductory Mechanics, four sessions per week- including teaching & grading).
- Taught undergraduate general Physics Lab (Electronics & Mechanics, 3 sessions per week- including lab equipment setup, guiding students through experiments, and explaining theoretical concepts).
- Teaching Assistant - Assisted professors with teaching classes, grading papers, & conducted workshops for assigned undergraduate/graduate Physics courses.

*Senior Laboratory Instructor*  
Department of Physics, UTEP

Jan 2018 - May 2019

- Assisted teaching assistants for undergraduate Electronics & Mechanics laboratory sessions (two sessions per week), providing support with laboratory training and experimental setup.

## **AWARDS & SCHOLARSHIPS**

- **Academic and Research Excellence Outstanding Graduate Student Computational Science**, UTEP (Dec 2024).
- **Graduate Research Award**, Graduate School, UTEP (Aug 2023- Aug 2024).
- **Best Oral Presentation Award**, New Mexico State University (NMSU)-NeSA 15th International Conference (Mar 2024).
- **Forum on Graduate Student Affairs (FGSA) URM March meeting award** (Feb 2021).
- **Reading is Fundamental (RIF) award**, College of Science, UTEP (Nov 2020).
- **Academic & Research Excellence Outstanding Graduate Student Physics**, UTEP (Dec 2019).
- **C. Sharp Cook Graduate Scholarship**, UTEP (Oct 2019).
- **Outstanding achievement: Better Rated by Students**, Physics, UTEP (May 2019).

## RESEARCH INTEREST

- Theoretical modeling, simulation, & visualization for understanding real-world phenomena & computation fundamentals.
- Develop a mathematical models using appropriate numerical methods, implement them in the form of a computer program, and visualize the numerical results.

## PUBLICATONS

- B. Ayirizia, **B. K C**, J. A. Muñoz, " *Physics-Informed Machine Learning for Refractory Alloy Design*". <https://doi.org/10.48550/arXiv.2608.03805> (ArXiv preprint).
- **B. K C**, R. Parajuli; " *First Principles Study of NaCl...A-B Type (A-B = Acceptor) Complexes*". Journal of Institute of Science and Technology, 30(1), 65–72. <https://doi.org/10.3126/jist.v30i1.68199>.
- **B. K C**, " *Thermal Anharmonicity of Transition Metals: A Case Study of Tantalum*" (2024). Open Access Theses & Dissertations. 4261. [https://scholarworks.utep.edu/open\\_etd/4261](https://scholarworks.utep.edu/open_etd/4261)
- C. Diaz-Caraveo, **B. K C**, & J. A. Muñoz; *Lattice Dynamics & Free Energies of Fe-V Alloys with Thermal & Chemical Disorder*. Journal of Physics: Condensed Matter. <https://doi.org/10.1088/1361-648X/ad66a5>.
- Homero Reyes-Pulido, Bimal K C, Ravhi S. Kumar, Russell J. Hemley, Jorge A. Muñoz; *Thermally frustrated phase transition at high pressure in B2-ordered FeV*. AIP Advances, 14 (7): 075108 (2024). <https://doi.org/10.1063/5.0219881>.
- S. Deng, **B. K C**, & V. Kreinovich; *Why Optimization Is Faster than Solving Systems of Equations: A Qualitative Explanation*. Uncertainty, Constraints, and Decision Making. Cham: Springer Nature Switzerland(2023). 341-344.
- **B. K C**, " *Quasi-Harmonic & Anharmonic Entropies in Transition Metals*" (2019). Open Access Theses & Dissertations. 2866. [https://scholarworks.utep.edu/open\\_etd/2866](https://scholarworks.utep.edu/open_etd/2866).

## UNDER REVIEW/ WORKING PAPERS

- **B. K C**, R. Ravelo, " *Anharmonic Contributions to Entropy and Thermal Pressure in Tantalum: Quasi-Harmonic Approximation versus Molecular Dynamics*".
- **B. K C**, J. A. Muñoz, R. Ravelo, " *Anharmonic Vibrational Entropy in Elemental Tantalum at High Temperature*".
- C. Garcia, **B. K C**, R. Ravelo, " *Comparative Study of Analytical Models of the Gruneisen Parameter of Metals as a Function of Pressure*".
- C. Diaz-Caraveo, D. A. Juarez, **B. K C**, E. O. Oyetunji, & J. A. Muñoz " *Effect of short-range order on the mechanical & thermal properties of shape-memory alloy NiTi*."

## CONFERENCE/ WORKSHOP PRESENTATIONS

- Space Exploration Educators Conference (SEEC), Space Center Houston, Houston, TX (Feb 4-7, 2026).
- " *First Principle Investigation of Magnetic, Elastic, & Thermodynamic Properties of Ordered D03 Fe<sub>3</sub>V*", New Mexico State University (NMSU) Nepalese Student Association (NeSA) 15th International Conference, Las Cruces, NM (Mar 16, 2024).
- " *Free Energy of the Order-disorder Phase Transition in FeV from Molecular Dynamics*", APS March Meeting, Minneapolis, MN (Mar 3 - 8, 2024).

- “*Harmonic Ensemble Lattice Dynamics of Crystals with Thermal & Configurational Disorder*”, 30th Joint NMSU/UTEP Workshop on Mathematics, Computer Science, & Computational Sciences, University of Texas at El Paso, El Paso, TXM (Oct 28, 2023).
- “*Why Optimization is Faster than Solving Systems of Equations: A Qualitative Explanation*”, 27th Joint NMSU/UTEP Workshop on Mathematics, Computer Science, & Computational Sciences, New Mexico State University, Las Cruces, NM (Apr 2, 2022).
- “*Anharmonicity in the Vibrational Entropy of Transition Metals*”, APS March Meeting, online (Mar 16, 2021).
- “*Classical Molecular Dynamical Simulations of Melting Curve of Copper*”, 10th International Conference, 2018, New Mexico State University, Las Cruces, NM (Mar 31, 2018).

**PROFESSIONAL SERVICE** Ad Hoc Manuscript Reviewer

- IOPScience - Machine Learning: Science and Technology (1)
- AIP Publishing - Physics of Fluids (2)
- IOPScience - Modelling and Simulation in Materials Science and Engineering (2)

**TEACHING EXPERIENCE**

Courses taught during the three most recent semesters  
**Fall 2026**

- University Physics I [PHYS 2425]
- College Physics I [PHYS 1401]
- Physical Science [PHYS 1415]

**Summer 2026**

- College Physics I [PHYS 1401]
- Physical Science [PHYS 1415]

**Spring 2026**

- University Physics II [PHYS 2426]
- College Physics II [PHYS 1402]
- Physical Science [PHYS 1415]

**EXPERTISE**

Atomistic modeling & simulations, High-Performance Computing (HPC), Quantum Computing, Data Mining, Cluster analysis, Machine Learning, Computational Statistics, Serial & Parallel Programming, etc.

**SOFTWARE SKILLS**

R, Python, Matlab, Mathematica, Gaussian, C(including OpenMPI), Java, SQL etc.

**SERVICE**

Nepalese Student Association at UTEP  
Vice President

Sep 2019 - Jun 2021